AI and Research Methods in Political Science

Large Language Models: Promise and Pitfalls

Valentina Gonzalez-Rostani
PhD Lecture

Overview: Six LLM Applications in Political Science

1. **Annotation & Measurement**
   Labeling and coding unstructured data

2. **Treatment Generation**
   Creating experimental stimuli

3. **Silicon Sampling**
   Synthetic survey respondents

4. **Agent-Based Models**
   Simulating social dynamics

5. **Tool Use & Web Browsing**
   Automated data collection

6. **LLMs as Objects of Study**
   Auditing AI systems

1 LLMs for Annotation & Measurement

Converting unstructured data into quantifiable categories

The Annotation Task

Traditional Workflow:

- Human researchers/RAs
- Codebooks & training
- Inter-coder reliability
- Expensive & slow

LLM Workflow:

- Prompt-based instructions
- Off-the-shelf models
- Scalable & fast
- Cost-effective

Example: "On a scale of 1-10, how positive is this text?
'It's a beautiful day!' → Only provide a number."

Cost Differences: A Dramatic Example

Task: Label sentiment of 1 million tweets
Codebook: 5,000 tokens | Tweet: ~50 tokens

Option A: Premium Setup

- GPT-5.2 with reasoning
- One tweet at a time
- No batch processing
- No prompt caching

Cost: $149,000

Option B: Optimized Setup

- GPT-5-nano, no reasoning
- 100 tweets per batch
- Batch API (50% discount)
- Prompt caching

Cost: $2.90

50,000× cheaper! Test performance vs. cost tradeoffs empirically.

Key Technical Decisions

Model Selection

- Open vs. closed weights
- Reproducibility tradeoffs
- Cost vs. performance

Temperature

- Set to 0 for reproducibility
- Higher for variation if needed
- Affects output variance

Reasoning

- Internal reasoning trace
- ~10× cost of input tokens
- Control via effort parameter

Input Batching

- Multiple cases per prompt
- 90%+ cost savings possible
- Must validate performance

⚠ **Document ALL choices, including defaults!**

Validation: Critical & Non-Negotiable

Why Validate?

- Performance varies by dataset/task
- LLMs can have systematic biases
- Errors may not be uniform
- Rare categories especially problematic

How to Validate?

- Compare to human-coded data
- Use out-of-sample evaluation
- F1 scores (not just accuracy!)
- Examine confusion matrices
- De-bias downstream analyses

Example: Hate Speech Detection

If LLM has 2% false positive rate but true hate speech is only 0.5% prevalent:
→ Model always choosing majority class performs well by accuracy but fails substantively

Always specify exact model version (e.g., gpt-5.2-2025-12-11, not just gpt-5.2)

Example: Political Advertisement Tone

Ornstein et al. (2025): Replicating Carlson & Montgomery (2017)

Original Study

- Mechanical Turk workers
- Coded ad tone
- Compared to expert codings
- Time & cost intensive

LLM Replication

- ChatGPT annotations
- Similar accuracy to MTurk
- Fraction of the cost
- Much faster

✓ Comparable quality to expert codings

⚠ But: Only validated for high-profile US politicians

Caution needed for: state legislators, foreign politicians, historical figures

2 LLM-Generated Treatments & Stimuli

Automating experimental design

The Experimental Design Challenge

Traditional Approach:

- Researchers create all stimuli manually
- Time-consuming and expensive
- Difficult to isolate causal variables
- Hard to create matched images/videos
- Training confederates is costly

LLM Promise:

✓ Generate many variants quickly
✓ More precise control over variation
✓ Reduce costs dramatically
⚠ BUT: Learned associations matter
⚠ Validation still critical

Multidimensional Treatment Problem:

Text varies in tone, length, word choice. Images vary in background, features.
→ How to isolate the causal dimension while holding others constant?

LLM Treatment Generation: Use Cases

1. **Static Treatments**

   AI-generated LinkedIn profiles varying race (Evsyukova et al., 2025)

2. **Text Generation**

   Novel propaganda items at scale (Goldstein et al., 2024)

3. **Personalized Content**

   Counter-attitudinal messages for each participant (Velez & Liu, 2025)

4. **Interactive Agents**

   LLMs as conversational partners for persuasion studies (Argyle et al., 2025)

Validation & the Bundled Treatment Problem

Key Concerns:

1. Construct Validity
   Does treatment vary only on target dimension?

2. Unintended Covariation
   Are non-target features changing too?

3. Scale vs. Validation
   Harder to check 100s of treatments

4. Disclosure Effects
   Participants knowing it's AI affects results

Best Practices

✓ Identify exact causal elements

✓ Record all LLM outputs

✓ Analyze outputs as data themselves

✓ Compare to human-generated when possible

✓ Pre-specify what should/shouldn't vary

3

Silicon Sampling

Synthetic survey respondents & digital twins

What is Silicon Sampling?

Using LLMs to generate synthetic proxies of individual-level research data

Basic Approach:

1. Construct a 'persona' with demographic/attitudinal characteristics
2. Prompt LLM to respond as that persona
3. Generate responses to survey questions or experimental stimuli
4. Create synthetic dataset for analysis

Motivations:

✓ Lower cost than surveys
✓ Instantaneous data
✓ Explore difficult/unethical questions
⚠ Rising LLM use by real panelists

Controversies:

⚠ Can LLMs truly replicate humans?
⚠ Validity of replacing humans
⚠ What are scientific standards?
⚠ Training data contamination

Critical Decision: Persona Construction

Example System Prompt:

```
"You are a 45 year old white male with a college degree, earning $75,000 per year.
You are a registered Democrat living in California in 2024."
```

Key Question: What information to include?

Unlike humans:

- Persona only exists via prompt
- Unlisted attributes are undefined
- More info → better performance
- But: risk over-weighting listed traits

Advanced approaches:

- Rich backstories from interviews
- Fine-tuning models on survey data
- Multi-hour depth (Park et al. 2024)
- Adjust model weights directly

Validation: Multiple Metrics for Different Goals

Individual Imputation	→	`F1, AUC, Cohen's κ`	*Predict specific person's response*
Distributional Simulation	→	`Means, variances, overlap`	*Replicate population patterns*
Relationship Estimation	→	`Coefficients, treatment effects`	*Causal inference & subgroup analysis*

⚠ **Choose validation metric based on your inferential goal!**

4 Agent-Based Models & Generative AI

From hand-coded rules to language-mediated interaction

Agent-Based Models: Core Concept

Computational simulations with autonomous agents that perceive, maintain state, and act

Traditional ABM Components:

1. Heterogeneous agent population
2. Interaction structures (networks, space)
3. Behavioral rules (hand-coded)
4. State update functions

Classic Example: Schelling's Segregation

Mild individual preference for same-group neighbors
↓
Extreme residential segregation emerges

→ Macro pattern ≠ simple aggregation of micro preferences

Generative Logic:

"Grow" collective phenomena from plausible micro-foundations rather than stipulating them.
Observe emergent patterns from repeated local interactions.

Generative ABMs (GABMs): The LLM Shift

Traditional ABM

- Hand-coded rules
- Scalar attributes
- Fixed heuristics
- Explicit equations
- Limited language
- Interpretable

GABM (LLM-based)

- LLM policy function
- Natural language personas
- Memory retrieval
- Black-box decisions
- Rich linguistic behavior
- Less interpretable

What changes:

✓ Can now model: persuasion, framing, norm contestation, threat signaling
⚠️ But: Less mechanistic transparency, replication challenges

Example: Simulating Social Media Dynamics

Törnberg et al. (2023): Twitter-like Platform Simulation

Setup:

- Agents initialized from ANES survey
- Generate posts, shares, engagement
- Vary ranking algorithms
- Measure toxicity from text

Interventions Tested:

- Engagement-ranked feed
- Global engagement ranking
- Cross-partisan bridging

→ Computational sandbox!

Other Applications:

- Whole-society simulations (Park et al. 2023: 25-agent town)
- Opinion dynamics & polarization (Chuang et al. 2024)
- Election simulations with campaign inputs (Zhou et al. 2025)
- Collective action cascades (Mou et al. 2024)

GABMs: Validation & Replication Challenges

Verification (Does it work as specified?):

✓ Sensitivity tests (prompts, temperature, order)
✓ Ablations (remove memory, reflection, tools)
✓ Cross-model robustness checks

Validation (Does it capture the phenomenon?):

✓ Match stylized facts & empirical benchmarks
⚠ Training data contamination checks
⚠ Avoid circular evaluation (LLM grading itself)

Replication Requirements:

- Full model specification (prompts, personas, memory, schedule)
- Exact model version & parameters
- Trace artifacts (transcripts, logs)
- Goal: Stable distributional patterns, not single narratives

5 Tool Use & Web-Browsing: LLMs as Research Agents

Automating data collection workflows

LLM-Driven Tool Orchestration

LLMs connected to external tools for finding, collecting, and organizing data

Traditional Approach:

- Manual web searches
- Copy-paste into spreadsheets
- Weeks/months for large datasets
- Custom scripts (fragile, technical)
- Human errors & inconsistency

LLM-Driven Approach:

- High-level instructions
- Automated search + extraction
- Seconds for complex tasks
- Handles PDFs, images, tables
- Flexible with messy data

Example Use Cases:

- Collect biographical data (e.g., education of all US Senators) → seconds
- Extract info from hearing transcripts (speakers, times) → automated
- Monitor policy documents over time → consistent & up-to-date

6

LLMs as Objects of Study

Auditing & evaluating AI systems

Three Motivations for Studying LLMs

1. Instrumentalist

Understanding LLMs to use them better as tools

Examples: Optimal hyperparameters, prompt engineering, task performance

2. Explanatory

Understanding how LLMs shape beliefs, behavior, and society

Examples: Bias detection, misinformation, labor market impacts, psychological harms

3. Aristotelian

Understanding LLMs as intelligent entities worth studying per se

Examples: Political opinions, human-like reasoning, preferences & beliefs

Auditing LLMs: Experimental Approach

Audit = Systematic evaluation of system behavior

Audit Experiment Design:

- Measure output Y (e.g., LLM response to ideology questions)
- Manipulate trait X (e.g., system prompt with different personas)
- Hold everything else constant
- Apply standard statistical tests: E[Y | X]

Example: Rozado (2023)

Applied standard ideology surveys to ChatGPT 3.5 → found left-leaning bias in 14 of 15 political orientation tests

⚠ Challenges:

- Converting text to structured data
- "Machine bias" (unexpected patterns)
- Prompt sensitivity (p-hacking risk)
- Must pre-register prompts!

Validity Challenges in LLM Audits

Construct Validity: Are we measuring what we think?

Instruments validated for humans may not work for LLMs
Example: LLM might choose 'A' because it's first alphabetically

Internal Validity: Do assumptions hold?

Do LLMs 'remember' prior treatments? SUTVA violations?
Training data contamination from published experiments?

Statistical Validity: What does variance mean?

LLM variance << human variance even at high temperature
Context-dependent: problem for synthetic surveys, not for bias measurement

External/Temporal Validity: Will results replicate?

Same prompt, same model, different date → different results
Rapid model updates make findings obsolete quickly

Cross-Cutting Themes

- Validation is always critical
- Reproducibility requires new standards
- LLMs expand what's possible but don't eliminate challenges
- Cost-performance tradeoffs matter
- Document everything (prompts, versions, settings)
- Beware: Rapid change = short shelf-life for findings

Broader Methodological Questions

- What happens when LLMs outperform humans? Death of the gold standard?
- How capable might LLMs become? We don't know the ceiling.
- Access barriers create equity problems in research.
- We don't fully understand how LLMs work (especially closed models).
- Should LLMs themselves be a central object of political science?
- What are appropriate standards for reproducibility with evolving systems?

Practical Recommendations for Researchers

Before Starting:

- ☐ Read relevant skills/documentation
- ☐ Test multiple models empirically
- ☐ Pilot on small sample first
- ☐ Pre-register prompts when possible

During Research:

- ☐ Document all decisions
- ☐ Record exact model versions
- ☐ Save all outputs & logs
- ☐ Validate, validate, validate!

For Publication:

- ☐ Share prompts & code
- ☐ Report all parameters
- ☐ Provide validation results
- ☐ Discuss limitations openly

For Reviewers:

- ☐ Check for validation evidence
- ☐ Assess reproducibility materials
- ☐ Evaluate prompt sensitivity
- ☐ Question cost-performance choices

Remember: LLMs are powerful tools, not magic. Critical thinking still required!

Conclusion

LLMs offer unprecedented opportunities for political science research:

- Faster & cheaper data collection and annotation
- New experimental designs previously impossible
- Synthetic data for exploration & piloting
- Complex simulations with language-mediated interaction

But challenges remain:

Validation • Reproducibility • Equity • Interpretability

The field is rapidly evolving. Stay critical. Stay rigorous.

Questions?

Valentina Gonzalez-Rostani